LECTURE AR ENHANCEMENT

HYPERFOCUS

TARGET AUDIENCE:

SCHOOL ENVIRONMENTS

INTERVIEW QUESTIONS FOR DEEPER UNDERSTANDING OF THE PROBLEM

Professor

Q1. In which school environments (lecture halls, labs, libraries, public spaces, etc.) do you feel students struggle most with engagement or understanding?

Q2. Are there specific teaching spaces or situations where enhanced visualization or interactive overlays could improve student learning?

Q3. Where do you see the current campus setup (classrooms, hallways, study areas) falling short in supporting effective instruction, and how might immersive or augmented tools help?

Students

Q1. In which parts of the school environment (classrooms, libraries, dormitories, dining areas, public spaces, commute) do you find learning or staying organized most challenging?

Q2. What kinds of support (visual explanations, real-time guidance, reminders, or interactive study aids) would help you in those specific locations?

Q3. How could XR tools (like AR overlays or immersive VR learning spaces) improve your learning experience or daily academic tasks in different school environments?

POV: Constructed From Interviews

01

Prof

Needs a way to keep students engaged and ensure key concepts are clearly understood during lectures.

02

Student A: ADHD Focus

Struggles to stay focused and often misses important information during class; needs visual cues and structured guidance.

03

Student B: Less English Proficient

Has difficulty understanding complex terminology and needs visual or multilingual support to follow lecture content.

POV: Prof

01

Prof

Needs a way to keep students engaged and ensure key concepts are clearly understood during lectures.

Attributes
Abilities: Strong subject expertise, efficient lecturer
Likes: Structured lessons, interactive demos, clear visuals.
Dislikes: Students disengaging, repeating instructions, low attendance

Goals
Deliver clear, engaging lessons that maintain student attention.

Need
Core requirement: A teaching environment that increases student engagement across different learning styles.

A professor needs to do: Quickly gauge comprehension, provide multimodal explanations, and keep students focused.
A professor wants to do: Enhance lessons without increasing workload
A professor does not: Does not have time to personalize explanations
A professor cannot do: Cannot monitor every student's engagement in real time during large lectures.

POV: Student A

02

Student A: ADHD Focus

Struggles to stay focused and often misses important information during class; needs visual cues and structured guidance.

Attributes
Abilities: Highly creative, strong problem-solving skills.
Likes: Interactive content, movement, visual cues
Dislikes: Long passive lectures, dense text, unclear instructions.

Goals
Stay attentive during class and retain information more effectively.

Need
Core requirement: A learning environment that naturally supports focus through structured cues

Student A needs to do: Manage attention and stay organized
Student A wants to do: Learn efficiently without feeling overwhelmed
Student A does not: Does not reliably stay focused in crowded environments.
Student A cannot do: Cannot sustain attention through long verbal instruction

POV: Student B

03

Student B: Less English Proficient

Has difficulty understanding complex terminology and needs visual or multilingual support to follow lecture content.

Attributes
Abilities: Strong analytical skills, high motivation, good visual learner.
Likes: Clear visuals, slow-paced explanations
Dislikes: Fast speech, dense academic language

Goals
Stay attentive during class and retain information more effectively.

Need
Core requirement: A learning environment that offers real-time language support, accessible explanations, and contextual clarification.

Student A needs to do: Translate on the fly and understand jargon quickly
Student A wants to do: Learn efficiently without feeling overwhelmed
Student A does not: Does not communicate clearly with professors
Student A cannot do: Cannot simultaneously translate and follow fast lectures

Needfinding Results

Professors / Instructors

- Students lose focus quickly during lectures
- Important points are not consistently understood
- Needs a tool that highlights key concepts and maintains engagement

ADHD/Students with Focus Difficulties

- Easily distracted and miss essential information
- Need visual cues or reminders to stay on track
- Benefit from structured, attention-guiding support

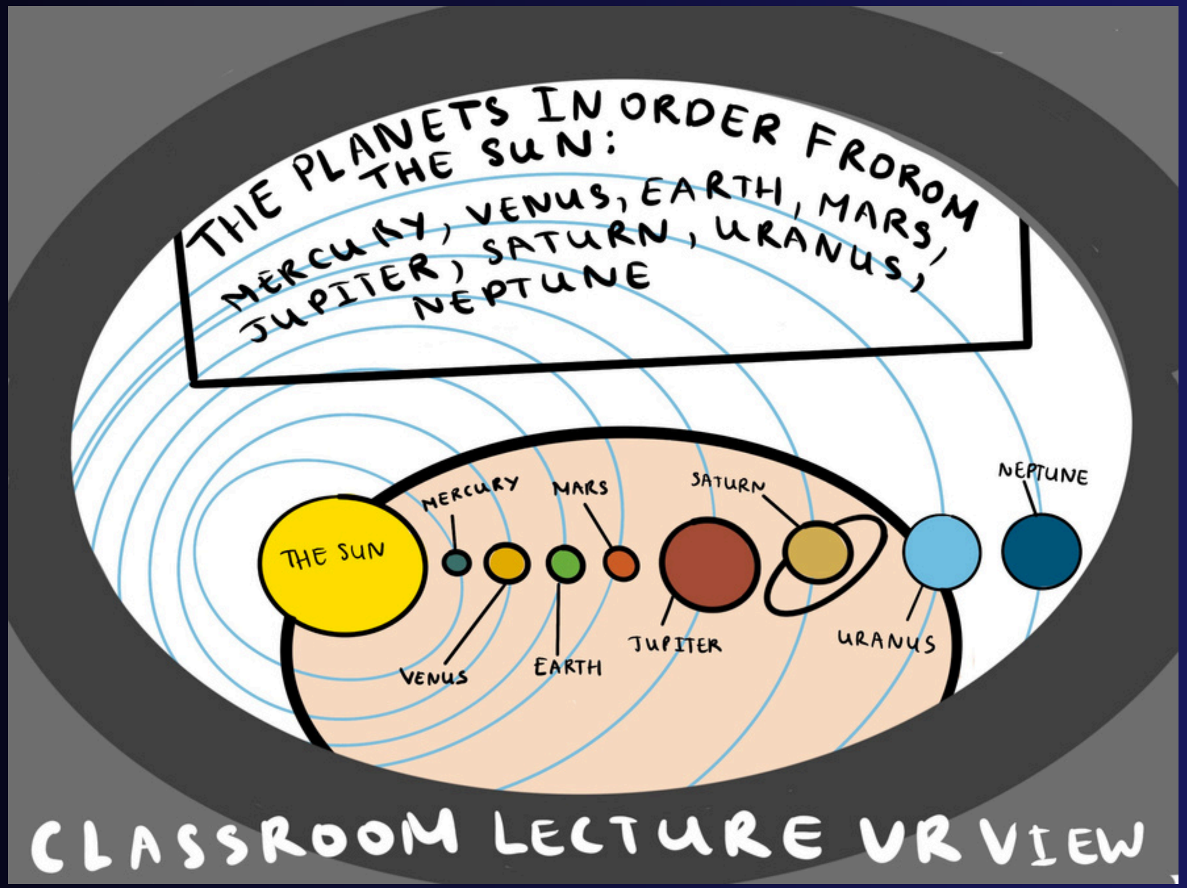
Low English Proficiency Students

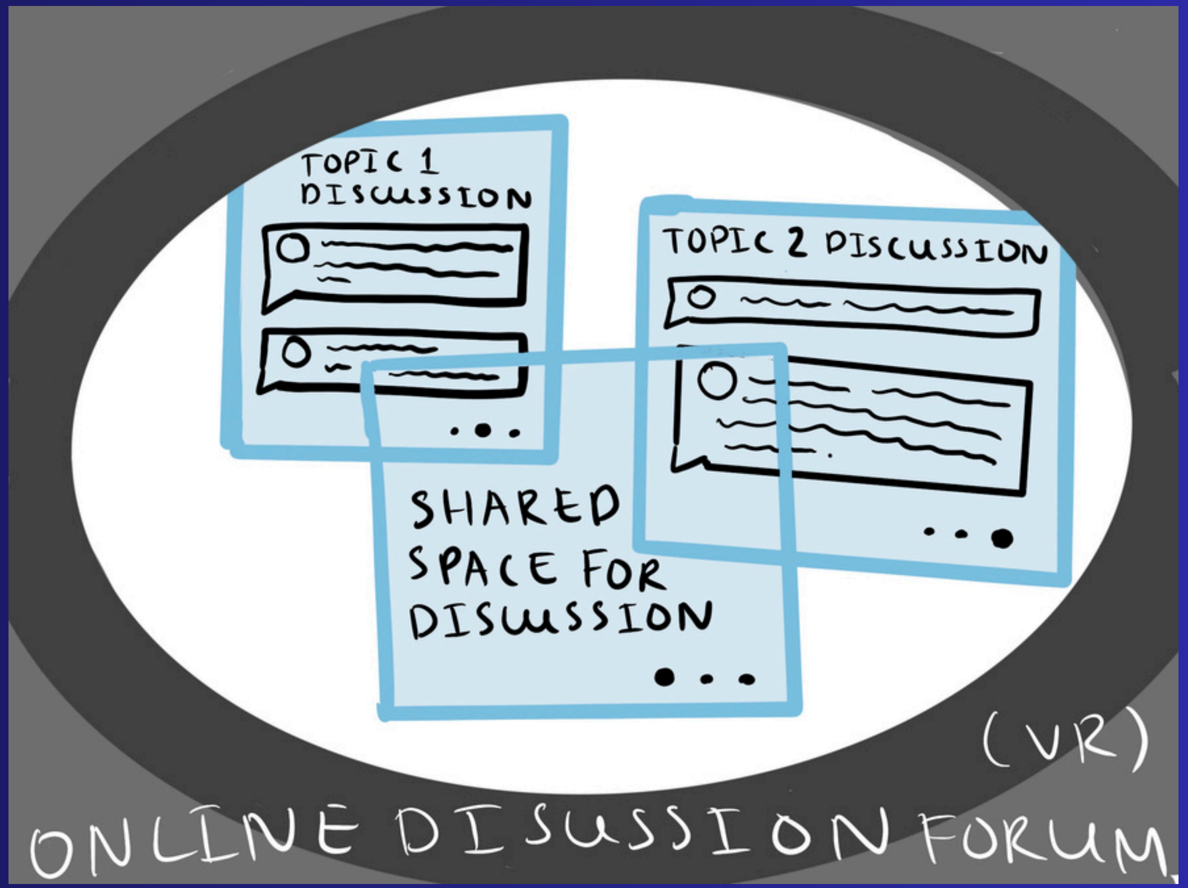
- Struggle with academic vocabulary and fast explanations
- Miss details due to language barriers
- Need multilingual or visual assistance to fully understand lessons

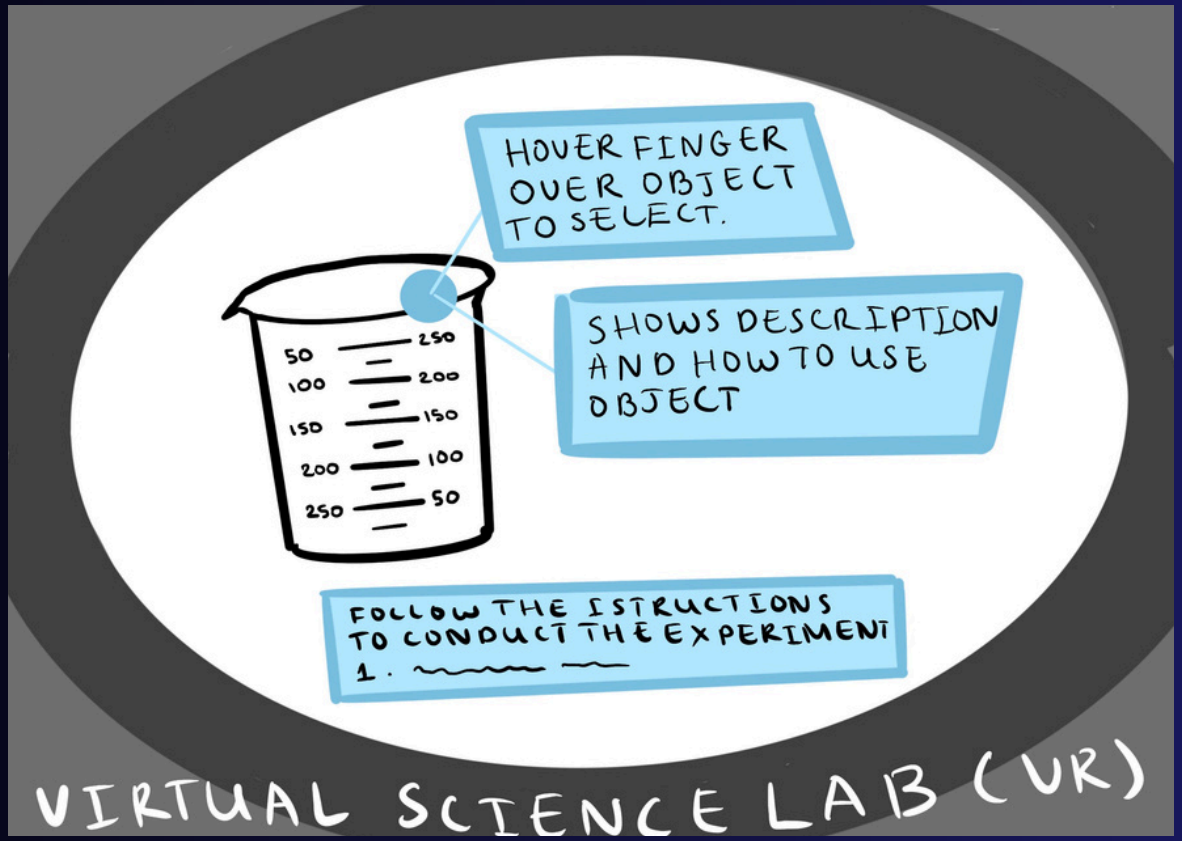
NARROWED TARGET:

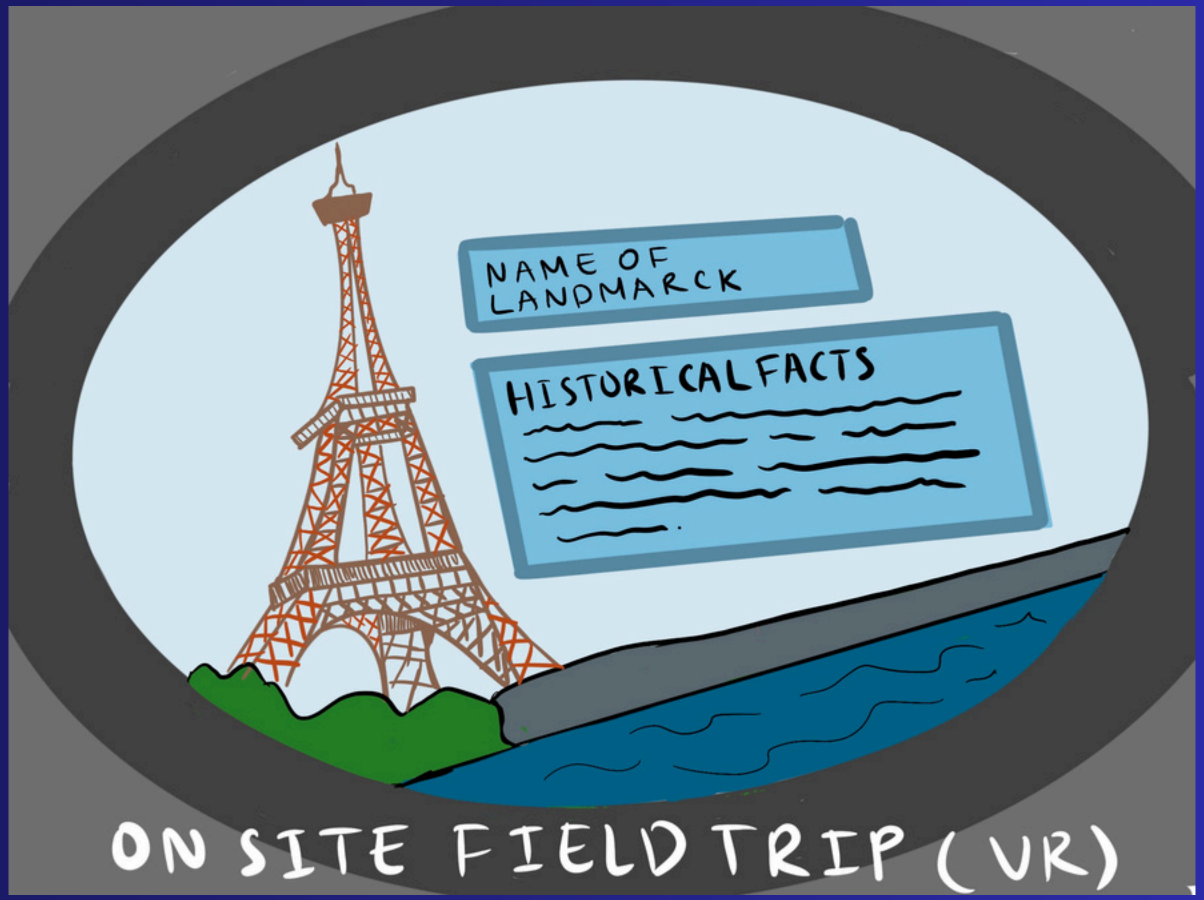
CLASSROOM LECTURE

STORYBOARDING

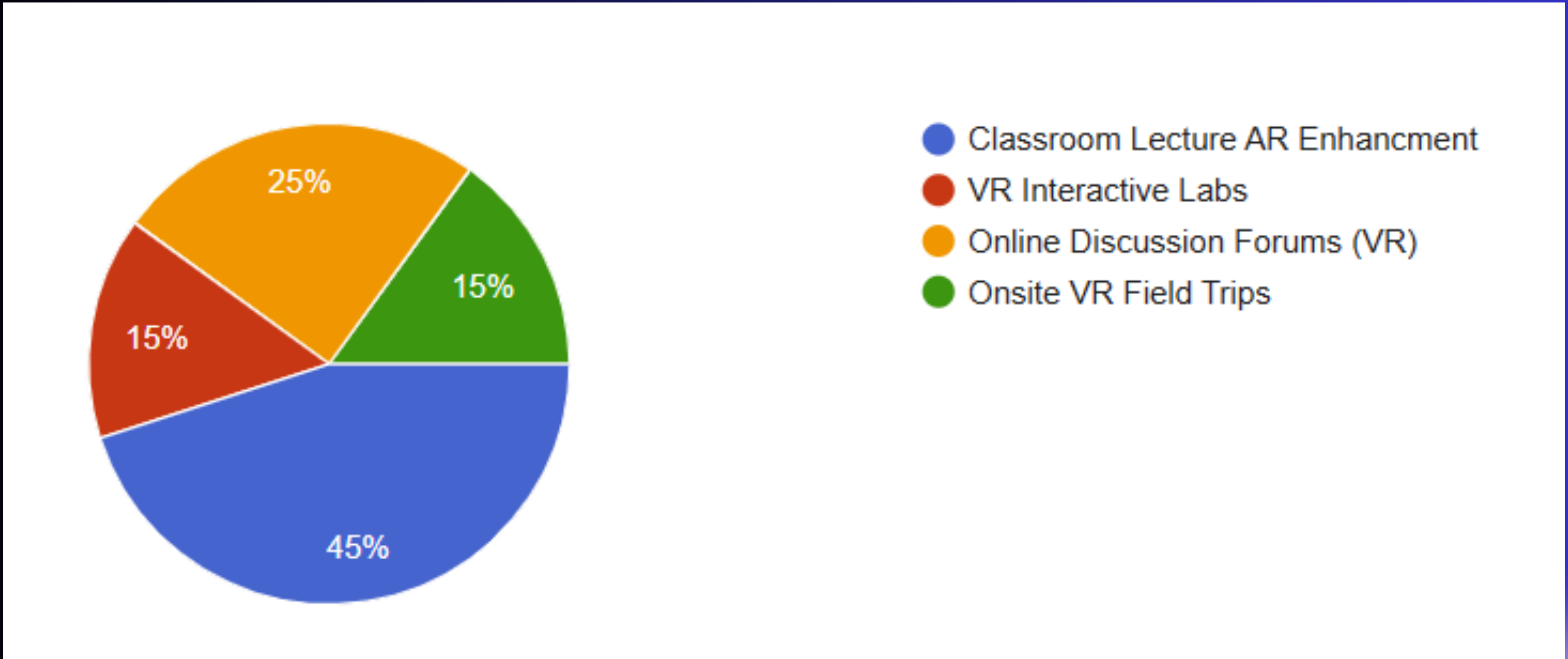




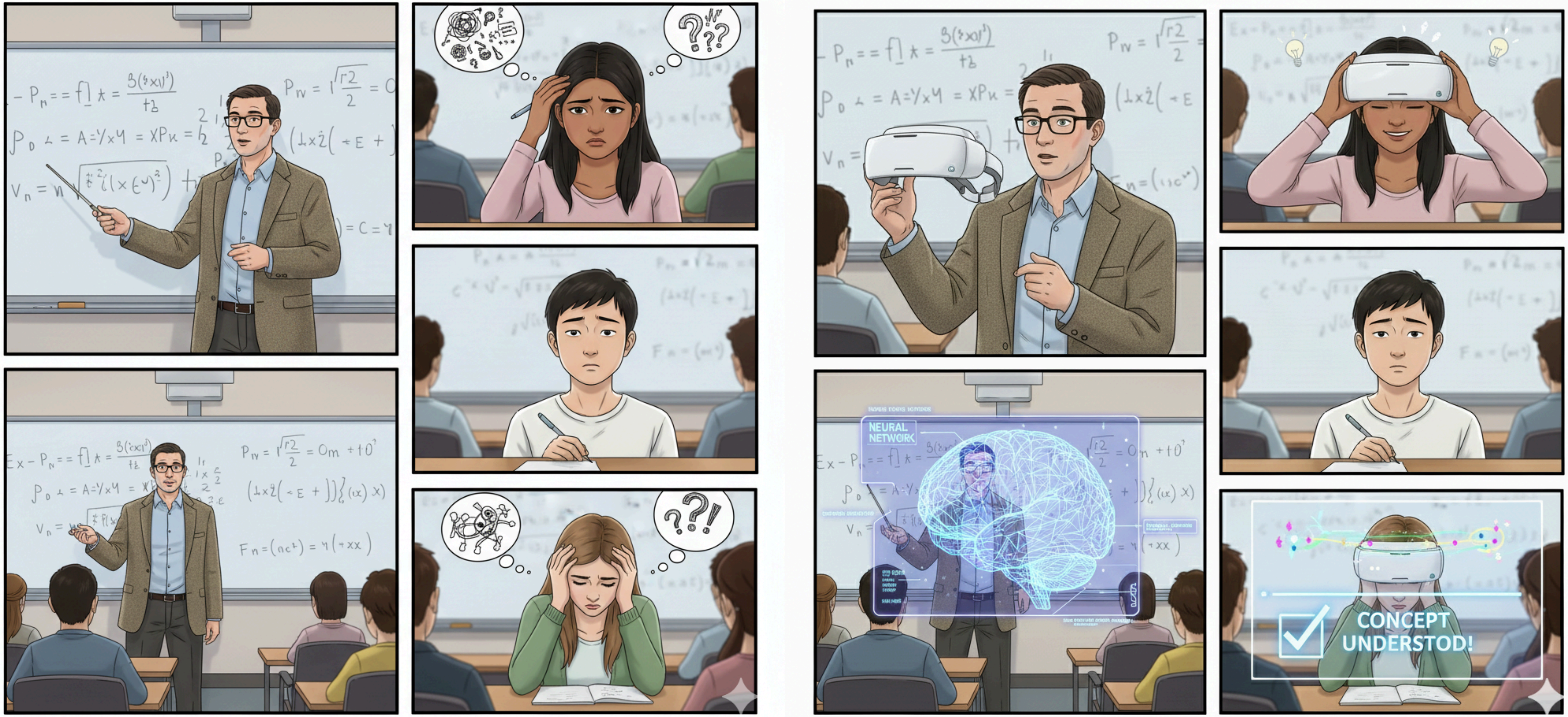




SPEED DATING EVALUATION

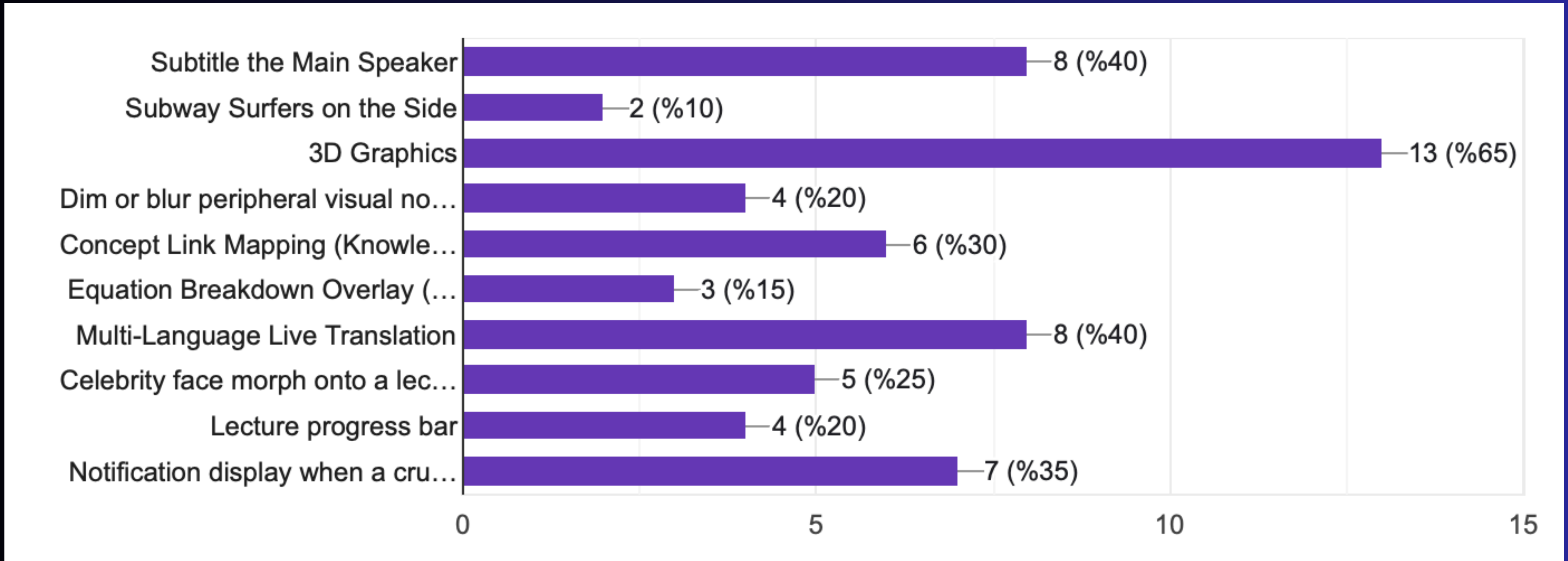


FINAL PROPOSED SOLUTION



FINAL PROPOSED SOLUTION

We propose an AR-based lecture enhancement system that combines the three most preferred features from our user research:

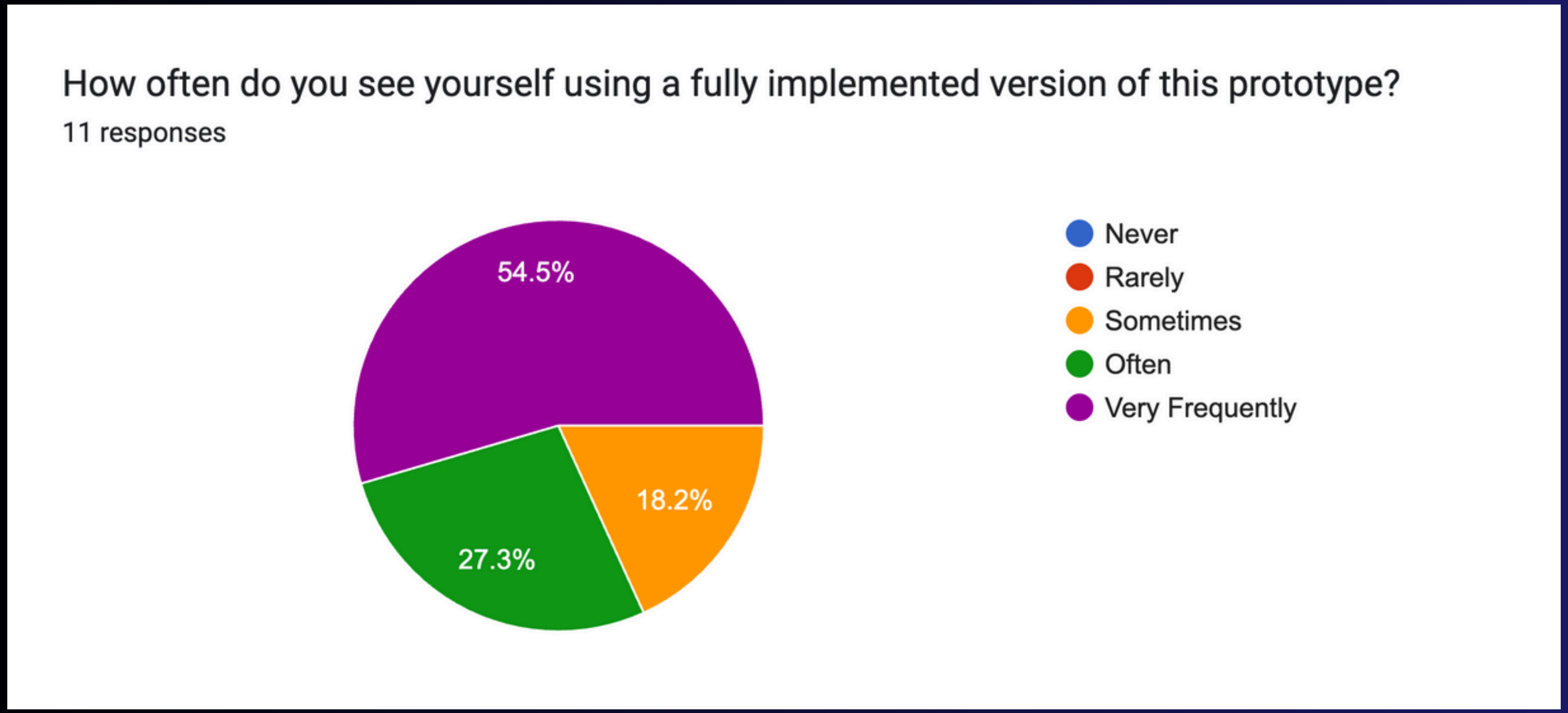


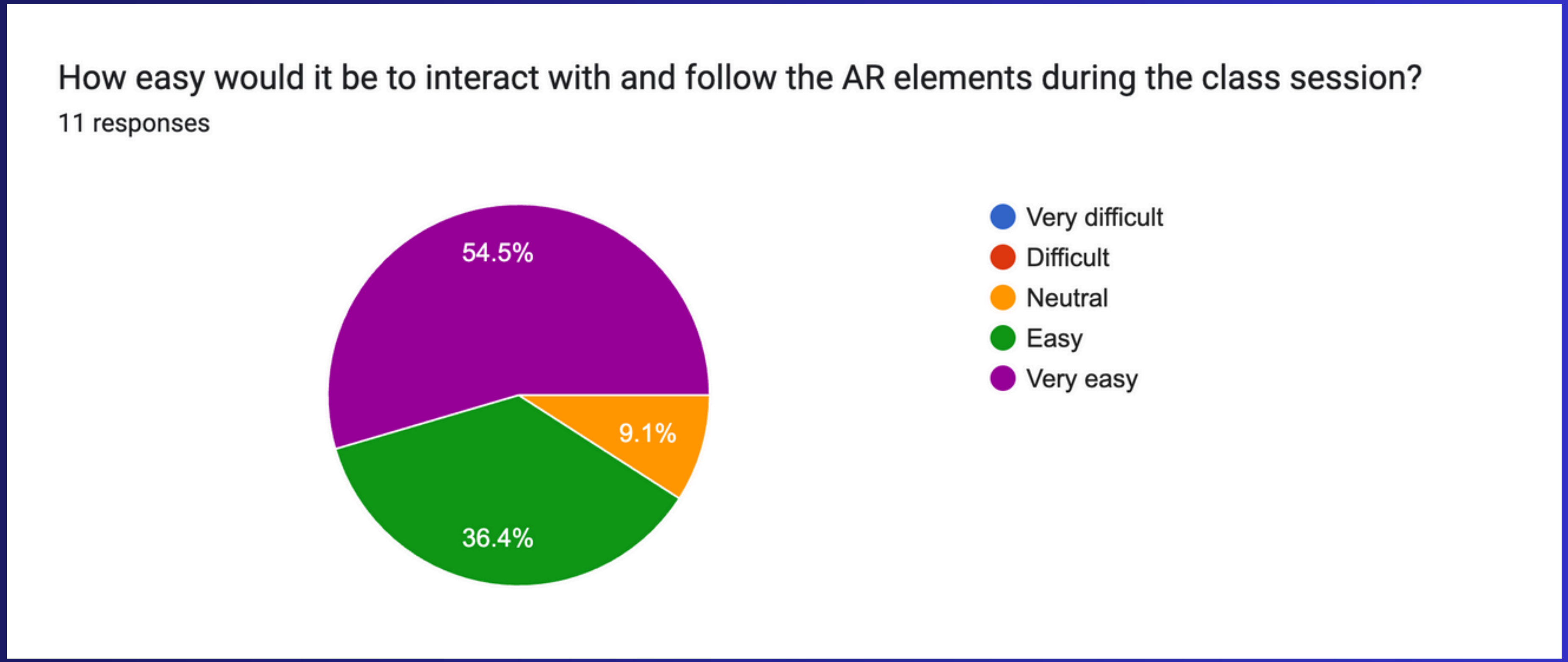
Together, these features help students better understand complex ideas, overcome language barriers, and stay engaged during lectures.

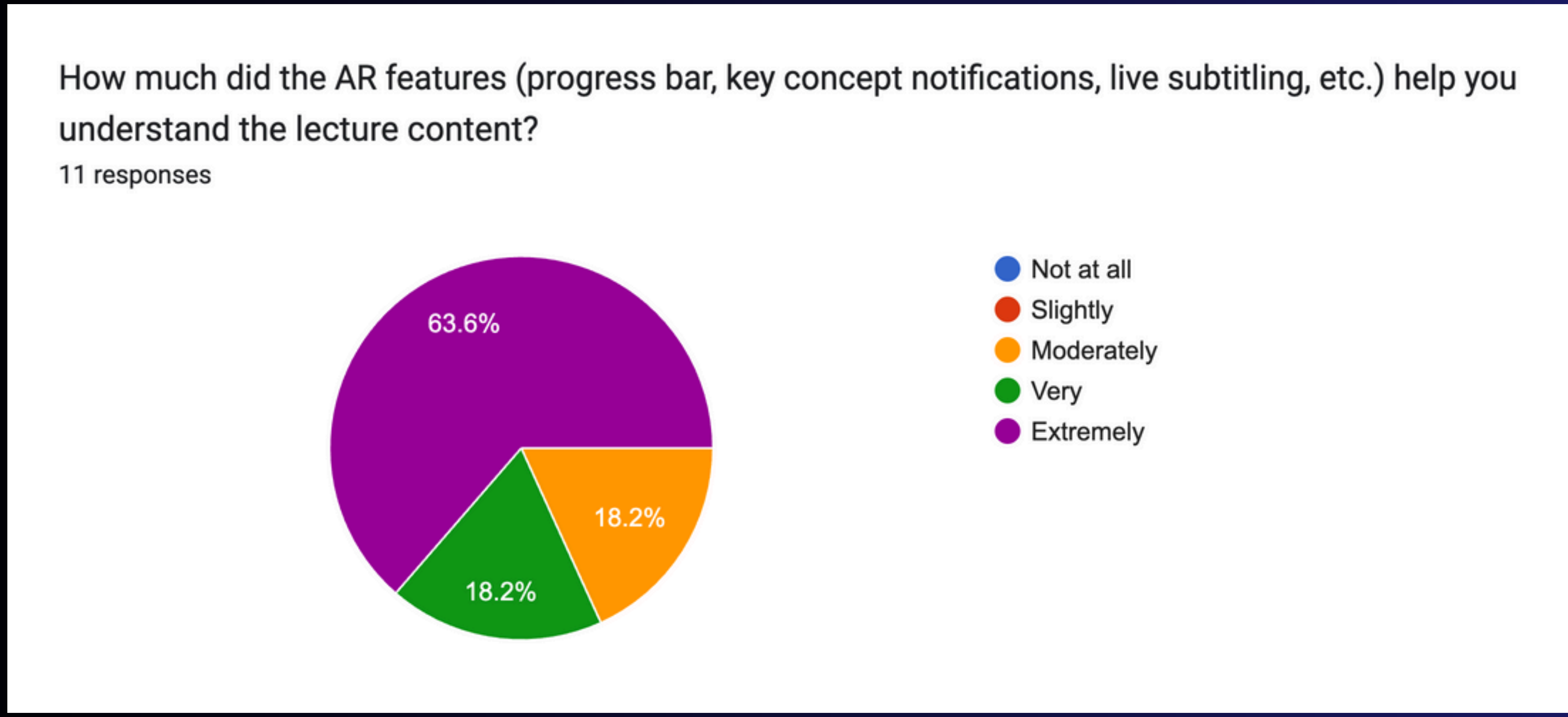
Prototype Demo

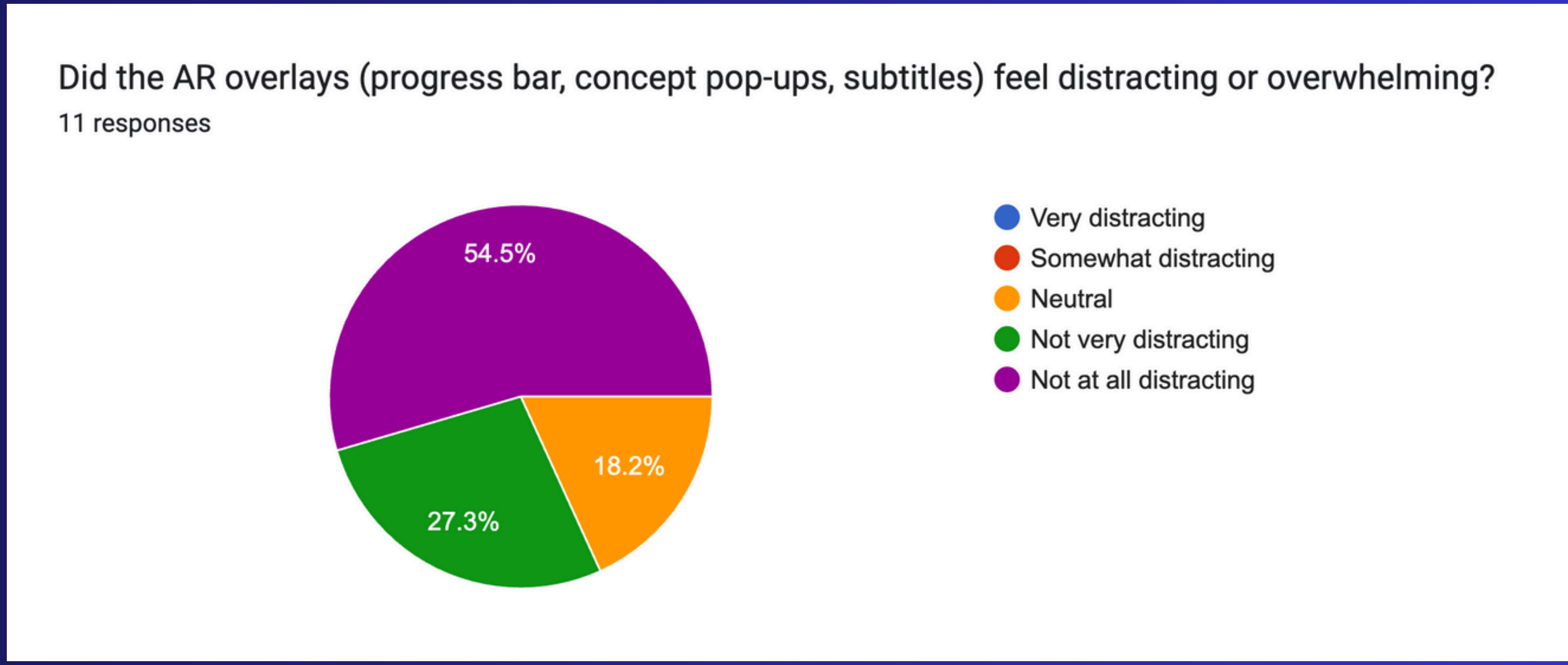
[Video link]

Prototype Evaluation









Reflection

Usability & Convenience

The AR elements were easy to follow and interact with during class, feeling smooth and intuitive.

Learning Enhancement

The features strongly boosted comprehension and made lecture content significantly clearer and more accessible.

THANK YOU

For Watching

Any questions?